FGE Infographic Compression Brief v1.0

Core Design Principle

> **Relative size communicates relative importance.** Every compression decision must preserve or reinforce visual hierarchy. A set of uniformly compressed infographics destroys the system. The hierarchy of the *set* is as important as the hierarchy *within* each piece.

Part 1 — FGE DNA Elements: Protection Matrix

These are non-negotiable. Aggressive compression destroys them quickly.

Priority	Element	Why It Matters	Compression Risk	Protection Rule
Critical	Material Signatures (Obsidian depth, Kintsugi gold seams, Pearl iridescence, Fracture lines)	Core visual language of the FGE universe	Color banding, crushed blacks/purples, loss of subtle gradients	Never push JPEG below 82 on material studies. Prefer PNG or lossless WebP for iridescence or thin-film effects.
Critical	Fine Technical Detail (atomic structures, dominance sliders, thin-film interference graphs, measurements, labels)	These are reference assets, not decoration	Loss of legibility on small text and fine lines	Primary and Secondary tier technical pieces must stay sharp. Text must remain readable at intended display size.

High	Visual Hierarchy & Importance Signaling	Core design principle — size = importance	Uniform softness flattens perceived importance	Maintain clear tier differentiation in final output sizes. Do not over-sharpen or over-smooth.
High	Kintsugi Gold & Fracture Lines	Philosophical and aesthetic signature of FGE	Gold seams turning muddy or flat	Protect warm gold tones in all tiers. Slight softness acceptable on Tertiary, but gold must still read as metallic and intentional.
High	Canon-Specific Systems (LangGraph architecture, Anchor Method, Haptic Veil Protocol, Emotional Instrumentation)	Operational doctrine — structural clarity is non-negotiable	Loss of diagram legibility under aggressive compression	These pieces default to Primary or Secondary tier regardless of original file size.
Medium	Color Temperature & Mood (crimson, obsidian purple, pearl shifts)	Emotional and material accuracy of the universe	Hue shift or desaturation	Run a color-accurate pass on any piece using non-standard palettes.
Low	Background texture / atmospheric effects	Supporting atmosphere	Can be softened more aggressively	Safe to compress harder on Tertiary. Does not carry canon information.

Part 2 — Pre-Work: Establish the Importance Tier Map

Before touching a single file, audit and rank all 30 infographics across three tiers. This map is your ground truth. Never compress without it.

Tier	Role	Target Display Size
Primary	Lead narrative, anchor pieces, canon doctrine	Largest — full-width or dominant
Secondary	Supporting context, detail, material studies	Mid — ~60–70% of Primary
Tertiary	Reference, supplemental, atmospheric	Small — ~35–45% of Primary

Automatic Tier Overrides (FGE Rules)

- Any infographic containing **canon doctrine** (LangGraph, Anchor Method, Haptic Veil, Emotional Vault, etc.) → **defaults to Primary**, regardless of original file size.
- Any infographic containing **both a hero visual and critical technical information** → **treated as Primary**, regardless of visual weight.
- **Material studies** (Obsidian, Blackrock, Pearl, Kintsugi) → **never below Secondary** unless purely decorative.

Part 3 — Compression Parameters

Format Decision Tree

Content Type	Preferred Format
Material signatures, iridescence, thin-film effects	PNG or lossless WebP
Photography-heavy	JPEG 82+ or WebP
Flat color / illustration / text-heavy	PNG-8 or WebP
Animation	WebP or optimized GIF
All outputs	Export a WebP version alongside primary format

File Size & Quality Targets by Tier

Tier	Max File Size	Min Resolution	Quality Floor	FGE-Specific Notes
Primary	450–650 KB	1800–2200px wide	JPEG 82+ or lossless WebP	Protect material signatures and fine technical detail at all costs. Use higher end of range for canon doctrine pieces.
Secondary	180–280 KB	1200–1400px wide	JPEG 78+ or WebP	Slight softness acceptable. No loss of gold seams or slider legibility.
Tertiary	60–110 KB	700–900px wide	JPEG 70–75 or aggressive WebP	Can be more decorative. Fine atmospheric detail can be sacrificed if hierarchy is preserved.

Part 4 — Hard Rules (Non-Negotiable)

1. **Always run lossless optimization first** before any lossy pass. Never skip this step.
2. **Never re-compress an already-compressed file.** Always work from the original master.
3. **Never compress material studies below Secondary tier** unless they are purely decorative.
4. **Never let text on technical diagrams become unreadable** — dominance sliders, atomic structures, specs, and measurement labels must remain legible.
5. **Primary tier canon doctrine pieces default to the higher end of the size range.**
6. **If an infographic contains both a hero visual and critical technical information, treat it as Primary** regardless of original file size or visual weight.
7. **Gold must still read as metallic and intentional** at every tier, including Tertiary.

Part 5 — Batch Workflow & File Structure

```
/masters/              ← originals, never touched
/compressed/
  /primary/
  /secondary/
  /tertiary/
/exports/
  /web/                ← WebP versions for all tiers
  /print/              ← high-res where required
```

Processing Order

1. **Primary tier first** — set your quality benchmark here against FGE DNA standards
2. **Secondary** — compress relative to Primary output quality, not the master directly
3. **Tertiary** — most aggressive pass last; verify gold still reads, hierarchy still holds

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Part 6 — File Naming Convention

```
[tier]-[sequence]-[descriptor].[ext]

Examples:
primary-01-langgraph-architecture.webp
secondary-04-kintsugi-material-study.png
tertiary-12-obsidian-atmosphere.webp
```

Rules:

- Tier prefix: `primary`, `secondary`, `tertiary`
- Sequence: zero-padded two-digit number
- Descriptor: lowercase, hyphenated, descriptive of content
- Extension: primary format first, WebP export alongside

Part 7 — QA Checklist (Per Infographic)

- Tier assignment confirmed against the Importance Tier Map
- Automatic tier override applied if canon doctrine or dual hero+technical content

- Text legible at intended display size
- No compression artifacts on gradients or fine lines
- Material signatures intact — Obsidian depth, Pearl iridescence, Fracture lines
- Kintsugi gold reads as metallic and intentional
- File size within tier target
- Aspect ratio unchanged
- Lossless optimization ran before lossy pass
- Filename follows convention: `[tier]-[sequence]-[descriptor].[ext]`
- WebP export generated alongside primary format
- Color temperature accurate (crimson, obsidian purple, pearl shifts not shifted or desaturated)

Part 8 — System-Level QA (After All 30 Are Processed)

View all 30 together at intended display sizes before sign-off.

- Does Primary tier read as visually dominant without any labels or context?
- Is the size step between tiers perceptible but not jarring?
- Do any Tertiary pieces accidentally compete with Secondaries?
- Does Kintsugi gold read as intentional at every tier?
- Is material signature integrity preserved across all Primary and Secondary pieces?
- Is the total page weight (sum of all 30) acceptable for the delivery context?

Part 9 — Anticipated Downstream Needs (n+3)

Responsive breakpoints — Tier sizes assume a single viewport. Define a secondary size for each tier at mobile (Primary collapses to 100vw, Secondary and Tertiary stack). Compress a mobile variant now while masters are open.

Dark mode variants — FGE's obsidian and deep-purple palette is dark-native. Any piece with a white or light background needs a dark-mode inversion pass. Cheaper to do now than after handoff.

Accessibility metadata — Map alt text to tier. Primary pieces require full descriptive alt text including material and canon references. Secondary pieces require descriptive alt. Tertiary atmospheric pieces can be `aria-hidden` if purely decorative. Define this before handoff.

Print resolution track — If any Primary or Secondary pieces will appear in physical or high-DPI print contexts, confirm 300 DPI versions exist in `/exports/print/` before the masters are archived.

FGE Infographic Compression Brief v1.0 — for use across all production lines.